SOP 2

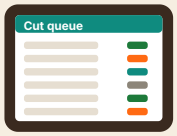
Load and cut a job

When: Every laser job (two 12 × 12 sheets)

Time: 2 to 3 min hands-on, then 15 to 25 min of attended cutting (Estimate) **Who:** Laser operator

You need: Planner tablet | Sheets for the job | Straight ruler | Honeycomb pins

1

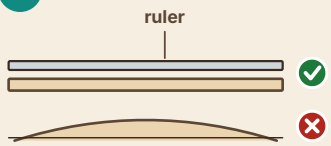


Pick the next job in the planner.

Cut queue lists the job file and the sheet colors. Oldest orders first.

- Example job: dragon_job3_E-F_x4.

2

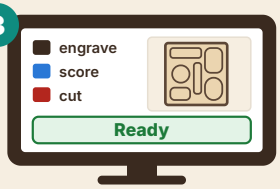


Check each sheet.

Lay it on the bench under a straight ruler. Color face up.

- Bowed more than 1 mm (Estimate): use it for the coupon or scrap.

3

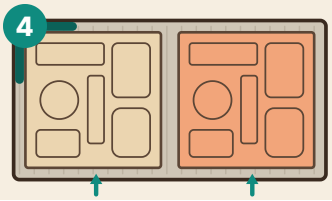


Open the job file.

`<job>_bed.lbrn2` in the kit's `cutfiles` folder. The order is set: engrave, score, holes, outlines.

- Speeds match the label on the machine.

4

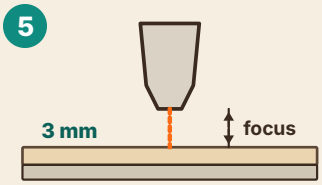


Load two sheets.

First sheet letter on the left, in the corner stop. Second sheet beside it, 5 mm gap.

- Flat on the bed. Pin a corner if it lifts.

5

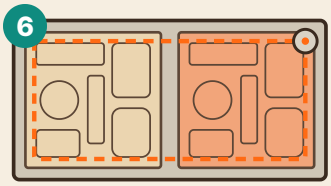


Set the focus.

Autofocus, or enter 3 mm material height.

- The nozzle does not touch the wood.

6



Frame the job.

Press Frame. The red dot traces the outline.

- The outline stays on both sheets, with margin all round.

7

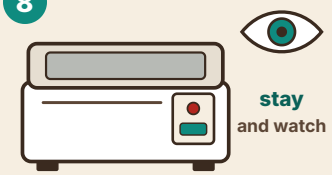


Close the lid and start.

Air assist and exhaust must be on.

- Cut lines look thin and dark, not wide and black.

8



Stay and watch the whole job.

Never leave the room while the laser runs, not even for a minute.

- A small flicker at the head is normal.

9

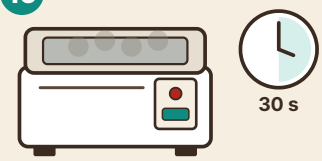


A flame stays after the head moves on? Press Pause.

Keep the lid shut. Most go out by themselves. If not: SOP-6.

- Log it as a burn on the Quality tab.

10

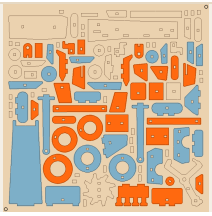


When it ends, wait 30 seconds.

Let the exhaust clear the smoke before you open the lid.

- Then unload with SOP-3.

11

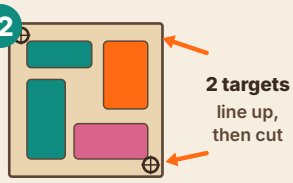


UV-printed color: print first.

Print the front, white and back files from the kit's `print_mode` folder.

- The print shows the 2 corner targets.

12



UV-printed color: Print and Cut.

In LightBurn, jog the red dot to target 1 and set it, then target 2, then align.

- The cut lands on the print edge. The 0.6 mm bleed hides small errors.

Planning settings (Estimate until you cut the coupon, SOP-5)

- Cut: 30 mm/s at 65% power. Score: 300 mm/s at 18%. Engrave: 400 mm/s at 20%, 0.1 mm line interval.
- After the coupon, write your own numbers on a label on the machine: cut ___ mm/s ___% kerf ___ mm.

Hartwood Designs, SOP-2, Rev A, 2026-10-12. Times and settings marked Estimate are planning numbers until measured on the real machine. Machine details follow the OMTech manual when it differs from this sheet.